BAYESIAN STATISTICS FOR PHD RESEARCHERS · AALBORG UNIVERSITY

Day 1 · Afternoon Probability and a first update

Random variables, a bag of marbles, and why the casino always wins

Lu Cheng and Ege Rubak · Department of Mathematical Sciences, Aalborg University

Monday 9 November 2026 · 12:30–16:00 · Pontoppidanstræde 101, room 1.001 · luch@math.aau.dk

THIS AFTERNOON

A language for uncertainty, then two games

I

Probability language

Events, conditional probability, random variables, PMF, PDF, CDF, iid — and the notation we use all week.

II

The marble game

What is in the bag? Our first use of Bayes' rule, one draw at a time.

III

Statistics in daily life

Is the casino's die loaded? And how does a casino win without cheating?

Then Exercise B: **your own timing data** — who is consistently short, who is long, and does feedback help?

I · PROBABILITY LANGUAGE

Outcomes, events, and four rules

Word	One die
Sample space Ω	{1, 2, 3, 4, 5, 6}
Event A	“even” = {2, 4, 6}
Probability P(A)	3/6 = 0.5 (fair die)

An **event** is a set of outcomes. A probability assigns each event a number.

The rules everything else follows from

$0 \le P(A) \le 1$

$P(\Omega) = 1$

$P(A \text{ or } B) = P(A) + P(B)$ if A and B cannot both happen

$P(\text{not } A) = 1 - P(A)$

Two readings, one set of rules: probability as a degree of belief in Bayesian statistics, e.g., “the proportion of blue marbles in this bag”.

I · PROBABILITY LANGUAGE

Conditional probability: restrict, then re-scale

DEFINITION

$$P(A \mid B) = P(A \text{ and } B) / P(B)$$

“Probability of A, given that we know B happened.”

PRODUCT RULE

$$P(A \text{ and } B) = P(A \mid B)\,P(B) = P(B \mid A)\,P(A)$$

ONE FAIR DIE

P(six | even) = (1/6) / (1/2) = **1/3**

P(even | six) = **1**

P(A | B) and P(B | A) are different things. Swapping them is the most common error in reading evidence — and Bayes’ rule is the tool for swapping correctly.

I · PROBABILITY LANGUAGE

A random variable: a number from a random process

X

Before you look

Capital letter. Not yet known; described by a **distribution**: which values, how probable.

x

After you look

Lower case. The value that actually came out: one number in your data.

Random variable	Possible values	Type
X = face of a die	1, 2, …, 6	discrete
Y = blue marbles in 4 draws	0, 1, 2, 3, 4	discrete
T = your next “10 seconds”	any value > 0	continuous

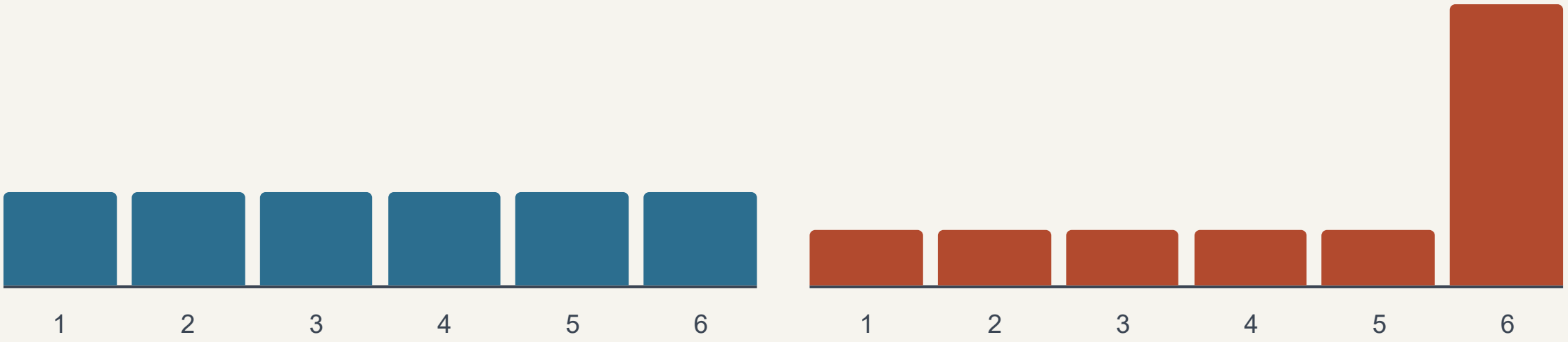
Discrete and continuous random variables

	Discrete	Continuous
Values	Countable: 0, 1, 2, …	An interval: any real number
Described by	PMF $p(x) = P(X = x)$	PDF $f(x)$, a density
Total is 1	$\sum_x p(x) = 1$	$\int f(x)\,dx = 1$
$P(X = x)$	Can be > 0	Always 0
Intervals	Add up the bars	Area under the curve
Examples	Die face, counts, yes/no	Time, weight, concentration
In R	dbinom, dpois	dnorm, dexp, dbeta

A PMF puts probability on each value: two dice

Fair die: p(x) = 1/6

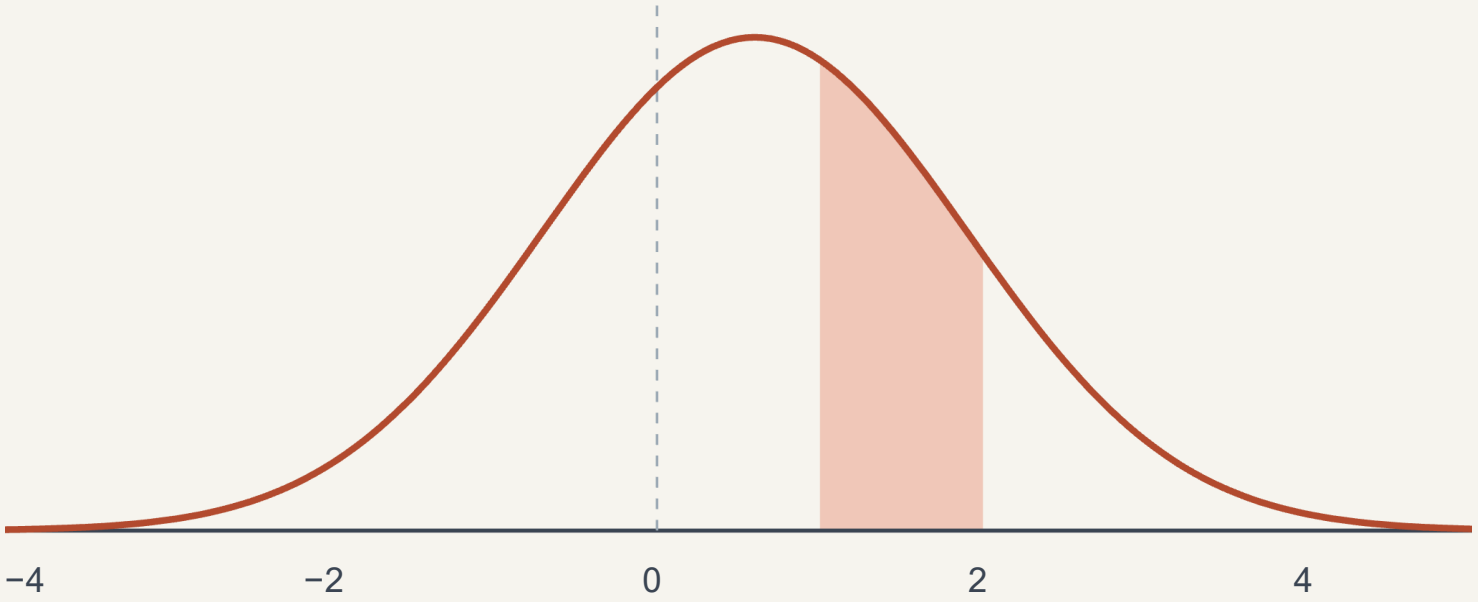
Loaded die: p(6) = 1/2, others 1/10



Both are valid PMFs: every bar ≥ 0, and the bars sum to 1. Keep the loaded die in mind — it returns in the casino.

I · PROBABILITY LANGUAGE · PDF

A PDF spreads probability out: areas are probabilities



Timing error (s) — an illustrative density, not real data

P(1 < T − 10 < 2) = shaded area

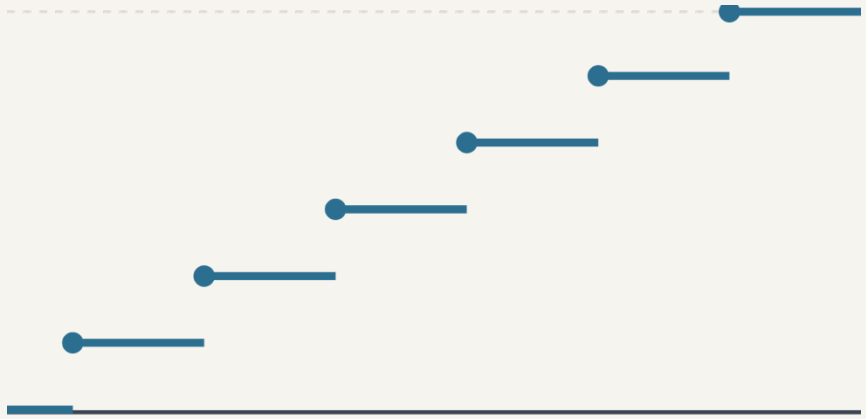
≈ 0.24

$P(a < X < b) = \int_a^b f(x)\,dx$

In R: **pnorm(2, 0.6, 1.3) − pnorm(1, 0.6, 1.3)**

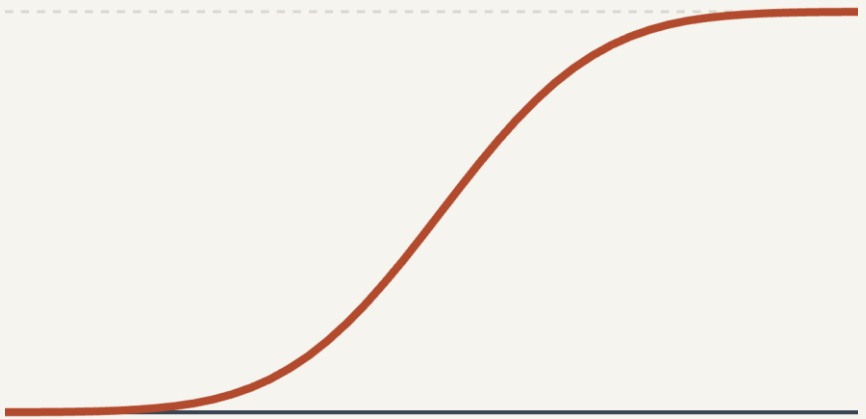
The CDF works for both kinds: F(x) = P(X ≤ x)

Fair die: a staircase



Jumps of 1/6 at 1, 2, …, 6 · pbinom, ppois

Timing error: a smooth S-curve



Rises steepest where the density is highest · pnorm, pexp

P(a < X ≤ b) = F(b) − F(a). The inverse of F gives quantiles: **qnorm(0.975)** = 1.96.

THE ONE THING TO REMEMBER ABOUT DENSITIES

A density is not a probability.

A density can be **larger than 1**. Uniform(0, 0.5) has f(x) = 2 for every x between 0 and 0.5 — the area is still 2 × 0.5 = 1.

It has **units**: probability per second. P(T = exactly 10.000 s) = 0; only intervals have probability.

I · PROBABILITY LANGUAGE · IID

iid: independent and identically distributed

Independent

Knowing one outcome tells you nothing more about the next — once the parameters are known.

Identically distributed

Every observation comes from the same distribution.

Written $Y_1, \ldots, Y_n$ **~ iid Bernoulli(θ)**. Marble draws **with replacement** are iid given the bag; without replacement they are not.

ARE YOUR 20 TIMING TRIALS IID?

Feedback may change your distribution · you may learn or tire across trials · different people differ. iid is an **assumption of a model**, not a property of the data.

Notation conventions

Symbol	Meaning	Marble game
Y, X	Random variable, before observing	next draw
y, x	Observed value	$y = 1$ (blue)
$y_1, \ldots, y_n$; n	The data ; sample size	B, W, B ; $n = 3$
θ (theta)	Unknown parameter	proportion blue
$Y \sim \text{Bernoulli}(\theta)$	“Y is distributed as …”	one draw
$p(\cdot)$, $f(\cdot)$, $F(\cdot)$	PMF, PDF, CDF (we often write p for all)	
$p(y \mid \theta)$	Data given parameter: likelihood	$\theta^2(1-\theta)$
$p(\theta)$, $p(\theta \mid y)$	Prior, posterior	before / after
$E[X]$, $\text{Var}(X)$	Expected value, variance	long-run average

I · PROBABILITY LANGUAGE · IN R

Four letters in front of every distribution

d

PMF or PDF value

p

CDF: P(X ≤ x)

q

Quantile: inverse CDF

r

Random draws

```
dbinom(2, size = 4, prob = 0.5) # P(Y = 2) = 0.375
pbinom(2, size = 4, prob = 0.5) # P(Y ≤ 2) = 0.6875
qnorm(0.975) # 1.96
rbinom(5, size = 4, prob = 0.5) # 5 simulated values
sample(1:6, 10, replace = TRUE) # 10 rolls of a fair die
```

II · THE MARBLE GAME · PLAYED LIVE

What is in the bag?

The bag holds **4 marbles**, each blue or white. You cannot look inside. Five bags are possible:



0 blue



1 blue



2 blue



3 blue



4 blue

The rule: a volunteer draws one marble, shows its colour, **puts it back**, shakes the bag. Repeat.

Before the first draw: which bag do you think it is? How sure are you?

II · THE MARBLE GAME

Name the unknowns before computing anything

Quantity	What it is	Values
θ	Proportion of blue marbles — fixed but unknown	0, 0.25, 0.5, 0.75, 1
Y_i	Draw i: 1 if blue, 0 if white — random, observed	0 or 1
Model	$Y_1, Y_2, \ldots \mid \theta \sim$ iid Bernoulli(θ)	$P(Y_i = 1 \mid \theta) = \theta$
Prior	Before any draw, every bag equally plausible	$p(\theta) = 1/5$ for each

Why iid? Because we replace and shake. Without replacement, a blue draw would change the next draw's chances.

Bayesian move: θ is not random in the bag — but **we** are uncertain about it, so we give it a probability distribution.

II · THE MARBLE GAME · BAYES' RULE

Turning p(data | bag) into p(bag | data)

$$p(\theta \mid y) = p(y \mid \theta)\, p(\theta) \,/\, p(y)$$

$p(\theta \mid y)$
Posterior: belief after the data

$p(y \mid \theta)$
Likelihood: how well each bag explains the data

$p(\theta)$
Prior: belief before the data

$p(y)$
Sum of the numerator over all θ: makes it add to 1

In words: posterior ∝ likelihood × prior. Multiply, then rescale to sum to 1.

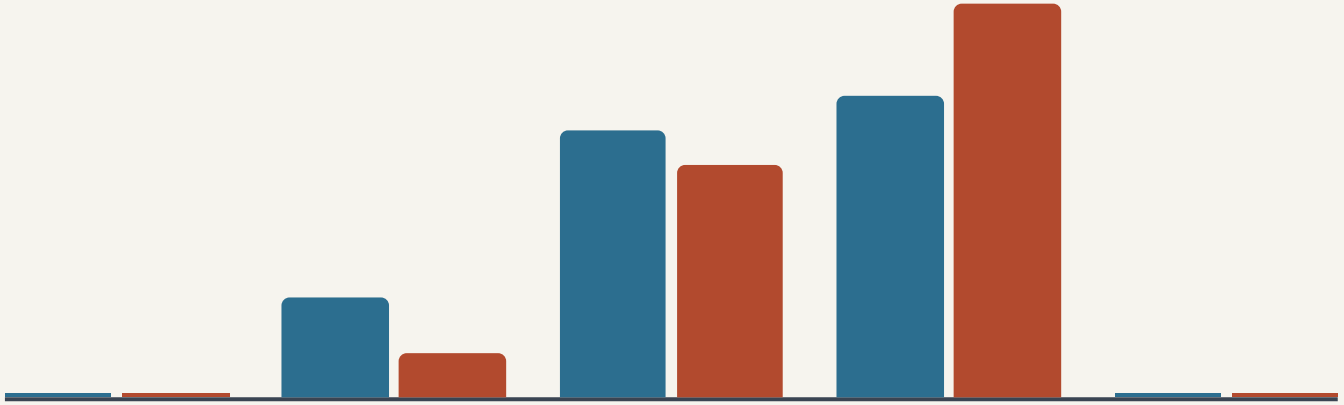
II · THE MARBLE GAME · WORKED EXAMPLE

We draw blue, white, blue. Which bag?

Bag	θ	Prior	Ways to get B W B (of 64)	Likelihood $\theta^2(1-\theta)$	Prior × lik.	Posterior
0 blue	0	0.2	0	0	0	0
1 blue	0.25	0.2	1 × 3 × 1 = 3	0.047	0.0094	0.15
2 blue	0.5	0.2	2 × 2 × 2 = 8	0.125	0.0250	0.40
3 blue	0.75	0.2	3 × 1 × 3 = 9	0.141	0.0281	0.45
4 blue	1	0.2	0	0	0	0
				Sum = p(y)	0.0625	1

Posterior = (prior × likelihood) / sum. One white draw rules out the all-blue bag; two blue draws rule out the all-white bag. “3 blue” is most plausible — but far from certain.

The fourth draw is blue: today's posterior is the new prior



Teal: after B W B · **Orange**: after one more B

UPDATE RULE

new posterior ∝
p(new draw | θ) × old posterior

Here: θ × (0, .15, .40, .45, 0), then rescale.

Same answer as using all four draws at once with likelihood $\theta^3(1-\theta)$. Order does not matter.

II · THE MARBLE GAME · PREDICTION

After B W B, what is the chance the next draw is blue?

TEMPTING: PLUG IN THE BEST BAG

0.75

Pretends we know it is the “3 blue” bag. We don’t: that bag has posterior 0.45.

BAYESIAN: AVERAGE OVER THE BAGS

0.575

$\sum \theta \cdot p(\theta \mid y) = 0.25 \cdot 0.15 + 0.5 \cdot 0.40 + 0.75 \cdot 0.45$

TRY IN YOUR GROUP · 3 MIN

(1) A friend packed the bag and says “at least two are blue”. New prior? New posterior? (2) What changes if we do **not** put the marble back?

III · STATISTICS IN DAILY LIFE · THE CASINO'S CHOICE

Cheat with the coin, or just keep you playing?

THE GAME

You start with **10 kr**. The casino tosses a coin; you guess heads or tails. Right: **+1 kr**. Wrong: **−1 kr**. At 0 kr you are out.

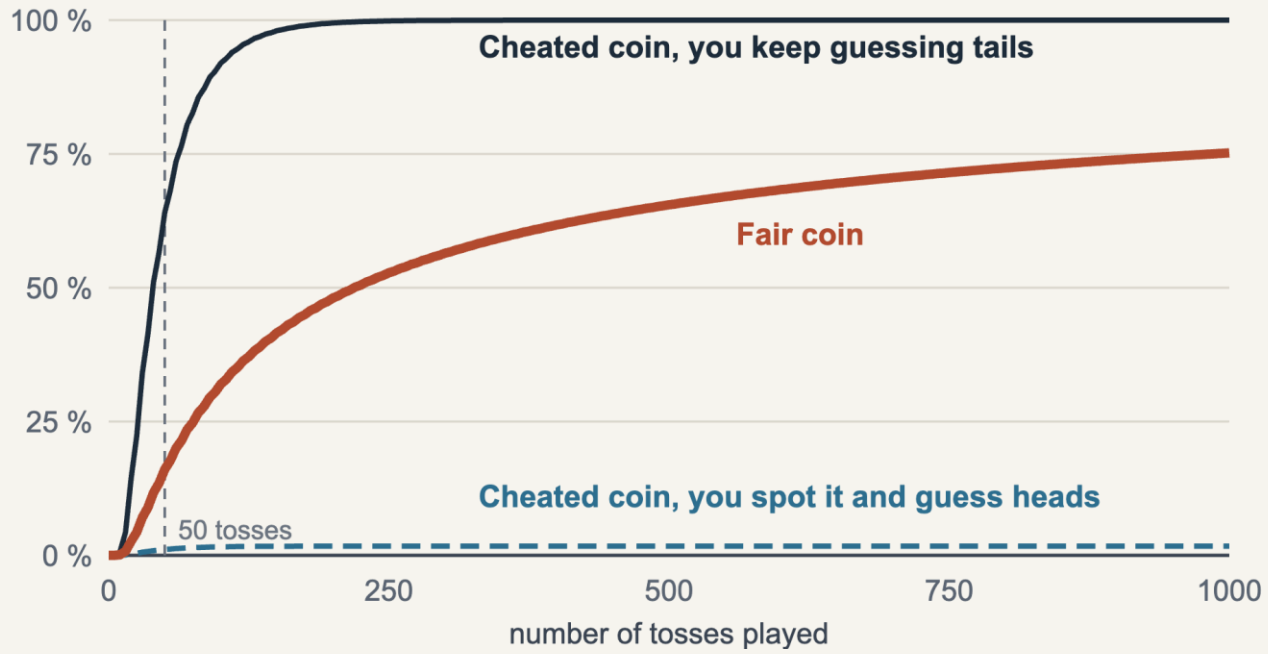
CHOICE 1 · A CHEATED COIN, 60 % HEADS

Wins only while you guess tails. A player who watches the tosses and updates (the marble game!) switches to heads — and then **beats the casino**.

CHOICE 2 · A FAIR COIN, AT LEAST 50 TOSSES

Nothing to detect, nothing to exploit. But the longer you play, the likelier your 10 kr runs out.

Probability you have lost all 10 kr by toss n



The casino picks the fair coin. With a fair coin you are broke with probability 16 % after 50 tosses, 48 % after 200 and 75 % after 1000 — and the casino has far more than 10 kr.

EXERCISE B · YOUR TIMING DATA

Five questions for your group

Q1 · BIAS

In the no-feedback trials, who in your group is **consistently shorter**, **consistently longer**, or shows **no clear difference**? How many of each?

Q2 · FEEDBACK

Does accuracy (mean |error|) improve with feedback — **for everybody** in your group? How many improved, how many got worse?

Q3 · PLOT

One figure: error by trial for each person, coloured by phase, with a line at 0. Compare it with the pooled group histogram.

Q4 · PREDICTION

Predict your own next no-feedback attempt as a range, and say where its uncertainty comes from.

Q5 · DENSITY VERSUS PROBABILITY

dnorm(0, mean = 0, sd = 0.2) returns 1.99. Is that a mistake? What is the probability that an error is exactly 0 s?

EXERCISE B · WHAT COUNTS AS “CONSISTENTLY”?

A working rule, and what chance alone would do

Rule for 30 no-feedback trials

21–30 short → consistently shorter
0–9 short → consistently longer
10–20 short → no clear difference

If you had no bias at all

Each trial is a coin flip: short ~ Binomial(30, 0.5). P(21 or more short) ≈ **0.021**. In 62 unbiased people, about **3** would get a label by chance.

```
d |>
  filter(condition == "no_feedback") |>
  group_by(participant) |>
  summarise(n_short = sum(short)) |>
  mutate(label = cut(n_short, c(-1, 9, 20, 30), labels = c("longer", "no clear", "shorter"))) |>
  count(label)
```

Use the same thinking for Q2: with no real effect, each person “improves” about half the time. Is 4 of 6 improving convincing?

EXERCISE B · 13:30–15:00

Your to-do list, in order

1 **Rotate roles**: new coder, checker, 2 question writers, interviewer, reporter.

2 Load your group's timing data; add **error_s**; check with **str()**.

3 **Q1**: classify each member (shorter / longer / no clear). Compare with your 08:45 predictions.

4 **Q2 + Q3**: mean |error| before vs after feedback per person, and one plot.

5 Q4 prediction and Q5 density question — short written answers.

6 Question writers: **concept, computation, pitfall**, each with a worked solution. One from what you found hard.

7 Checker re-runs the code and checks the solutions.

8 Reporter: note your group's hardest open question, to ask in the plenary.

Everyone back in the room at 15:00 for Peer review B. Take a break when it suits your group.

PEER REVIEW B · 15:00–15:45

Ask your partner group

YOUR PARTNER GROUP · THE SAME EVERY SESSION

1 ↔ 2 · 3 ↔ 4 · 5 ↔ 6

7 ↔ 8 · 9 ↔ 10

TWO ROUNDS · NOBODY MOVES

Round 1 (10 min): the odd group asks the even group.

Round 2 (10 min): swap.

Turn your chairs to the neighbouring table.

Time	Step
15:00–15:20	Two rounds — two or three questions each, to named people
15:20–15:43	Plenary: two groups drawn · your open questions
15:43–15:45	Wrap-up

DAILY DIARY · 15:45–16:00 · INDIVIDUAL

Before you leave: three and three

3

things I learned today

e.g. why P(A | B) ≠ P(B | A); how one draw updates the bag

3

things I am not yet familiar with

Be specific: “why a density can exceed 1” helps more than “PDFs”.

Go to **menti.com**, enter code **[code]** and type your answers there — no file, nothing on Moodle. Tomorrow’s first five minutes answer your “not yet familiar” lists.

TOMORROW · DAY 2

From five bags to infinitely many

θ can be any number between 0 and 1: Bayesian updating with the beta–binomial model, credible intervals, and prediction.

Tuesday 10 November · 08:45 · **Kroghstræde 3, room 4.110** (a different building) · nothing to prepare